Codebook for

"What We Think Others Think and Do About Climate Change:

A Multi-Country Test of Pluralistic Ignorance and Social Consensus Messaging"

Study 1

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Project description: In this project, we assessed climate change beliefs and perceptions thereof as well as the effectiveness of disclosing the actual beliefs on several factors related to climate action. We collected cross-stratified samples in terms of age and sex across 11 countries (raw data N = 8,151; cleaned data N = 3,653; Brazil, Canada, China, Germany, India, Indonesia, Italy, Japan, Mexico, Poland, and Thailand). The data were collected through respondi, an external panel provider certified under ISO 20252, from September 14 to October 13, 2022.

Table 1 provides core information for each variable in the cleaned data files:

- *data.csv / data.rds:* contains the cleaned datasets from all 11 countries
- *data.H1.csv / data.H1.rds*: to test H1 and RQ1 concerning pluralistic ignorance
- *data.int.csv / data.int.rds*: to test H3, H4a, H4b, H5a, and H5b concerning the intervention's effectiveness (including only climate change believers)
- *data.int.RQ2.csv / data.int.RQ2.rds*: to test RQ2 concerning the intervention effect on group efficacy beliefs (only including climate change believers and excluding those observations that have missing values on group efficacy beliefs)
- *data.exp.csv / data.exp.rds*: for exploratory analyses (only climate change believers who underestimated the number of climate change believers prior to being exposed to the intervention)

Table 2 provides core information for each variable in the YouGov Globalism dataset.

Table 3 provides core information for each variable in the 11 raw data files:

- *Brazil.csv*
- *Canada.csv*
- *China.csv*
- *Germany.csv*
- *India.csv*
- *Indonesia.csv*
- *Italy.csv*
- *Japan.csv*
- *Mexico.csv*
- *Poland.csv*
- *Thailand.csv*.

After downloading these datasets from Qualtrics, we delete two variables, "ResponseID" and the panel ID "m", to ensure anonymization.

Table 1

Core Information for Each Variable in the Cleaned Data Files Used for the Analyses (data.csv/.rds, data.H1.csv/.rds, data.int.csv/.rds, data.int.RQ2.csv/.rds, data.exp.csv/.rds)

Variable category	Variable name	Variable label and item wording	Variable type	Valid values	Value labels	Missing values
Collected by Qualtrics	recaptcha	Google's invisible reCaptcha for bot detection	numeric	0-1	> 0.5 = participant is likely a human ≤ 0.5 = participant is likely a bot	—
General	ID	Unique participant ID	numeric	≥ 1	—	—
	consent	Consent	numeric	1	1 = agreed	—
	response.type	Response type	character	Complete	Complete = complete and valid response	—
	dur	Duration of survey completion (in seconds)	numeric	≥ 0	—	—
	comp	Whether participant passed or failed the comprehension check	numeric	0, 1	0 = failed check 1 = passed check	—
Demographics	citizen	Are you a citizen of [country]?	numeric	0, 1	0 = no 1 = yes	—
	residency	Do you currently live in [country]?	numeric	0, 1	0 = no 1 = yes	—
	country	Assigned when merging the datasets	character	Brazil, Canada, China, Germany, India, Indonesia, Italy, Japan, Mexico, Poland, Thailand	—	—
	age (in years)	How old are you?	numeric	0, ≥ 18	0 = younger than 18	—
	sex	What is your sex?	numeric	0, 1	0 = female; 1 = male	—

	age.sex	Combined variable of age and sex used for reweighting	character	18-29f, 30-39f, 40-49f, 50-59f, 60f, 18-29m, 30-39m, 40-49m, 50-59m, 60m	18-29f = 18-29 years, female 30-39f = 30-39 years, female 40-49f = 40-49 years, female 50-59f = 50-59 years, female 60f = 60 years or older, female 18-29m = 18-29 years, male 30-39m = 30-39 years, male 40-49m = 40-49 years, male 50-59m = 50-50 years, male 60m = 60 years or older, male	—
	edu	What is your highest level of completed education?	numeric	1-7	Adapted to each country 1 = lowest (no formal degree) 7 = highest (doctoral degree)	—
	pol	In politics, people sometimes talk of “left” and “right”. Where would you place yourself on this scale, where 0 means the left and 10 means the right?	numeric	0-10	0 = left 10 = right	—
	urban	Would you describe the area where you live as urban or rural?	numeric	1-3	1 = urban 2 = rural 3 = don’t know/prefer not to say	—

National identification	ident1	I identify as [nationality].	numeric	0-10	0 = strongly disagree 5 = neither disagree nor agree 10 = strongly agree	—
	ident2	Being a [nationality] is an important reflection of who I am.	numeric	0-10	0 = strongly disagree 5 = neither disagree nor agree 10 = strongly agree	—
Own and others' climate change beliefs	own.ccb	In general, which of the following statements, if any, best describes your view?	numeric	1-5	1 = The climate is changing and human activity is mainly responsible. 2 = The climate is changing and human activity is partly responsible, together with other factors. 3 = The climate is changing but human activity is not at all responsible. 4 = The climate is not changing. 5 = I don't know.	—
	other.ccb.main	What percentage of [citizens] do you believe, would think the following ways about climate change? Please indicate a number from 0% (no one) to 100% (everyone) for the following statements such that they sum up to 100%. The climate is changing and human activity is mainly responsible.	numeric	0-100	—	—

	other.ccb.part	The climate is changing and human activity is partly responsible, together with other factors.	numeric	0-100	—	—
	other.ccb.attr	The climate is changing but human activity is not at all responsible.	numeric	0-100	—	—
	other.ccb.trend	The climate is not changing.	numeric	0-100	—	—
	other.ccb.know	Don't know. Please indicate what percentage of [citizens] do not know what they think about climate change.	numeric	0-100	—	—
	own.ccb.believer	Whether participant believes in human-caused climate change (response 1 or 2 on own.ccb) or not (response 3, 4, or 5 on own.ccb)	numeric	0, 1	0 = non-believer 1 = believer	—
	other.ccb.believer	Sum of other.ccb.main and other.ccb.part for each participant	numeric	0-100	—	—
Outcomes	discuss	How willing or unwilling are you to express your opinion on climate change among [citizens] you don't know?	numeric	1-7	1 = not at all willing 7 = very willing	—
	exp.change.low	What percentage of [citizens] do you believe, would be willing to make the following extent of changes to how they live and work to help reduce the potential effects of climate change? Please indicate a number from 0% (no one) to 100% (everyone). No or a few changes	numeric	0-100	—	—

	exp.change.high	Some or a lot of changes	numeric	0-100	—	—
	own.change	How much, if anything, would you be willing to change about how you live and work to help reduce the potential effects of climate change?	numeric	1-4	1 = no changes 2 = a few changes 3 = some changes 4 = a lot of changes	—
	exp.support.low	What percentage of [citizens], do you believe, think climate change should be a low/medium or high/very high priority of the government of [country]? Please indicate a number from 0% (no one) to 100% (everyone). Low or medium priority	numeric	0-100	—	—
	exp.support.high	High or very high priority	numeric	0-100	—	—
	own.support	Do you think climate change should be a low, medium, high, or very high priority for the government of [country]?	numeric	1-4	1 = low priority 2 = medium priority 3 = high priority 4 = very high priority	—
	efficacy	To what extent do you think that [citizens] can jointly prevent the negative consequences of climate change?	numeric	1-7	1 = not at all 7 = very much 999 = I don't believe in (human-caused) climate change. NA = I don't know.	—
Other	cond	Control or intervention condition	numeric	-0.5, 0.5	-0.5 = control 0.5 = intervention	—
	cc.first	Whether participant was first presented with the block on climate change vs. biodiversity loss	numeric	-0.5, 0.5	-0.5 = biodiversity first 0.5 = climate change first	—
	own.first	Whether participant was first presented with the block on own vs. other beliefs	numeric	-0.5, 0.5	-0.5 = other beliefs first 0.5 = own beliefs first	—

	order	Randomization of response options of own climate change beliefs	numeric	-0.5, 0.5	-0.5 = 4|3|2|1|5 = option 1 first, option 5 last 0.5 = 1|2|3|4|5 = option 4 first, option 5 last	—
	weights	Post-stratification weights calculated using automated raking based on the joint distribution of age and sex as well as the marginal distribution of climate change beliefs from the YouGov Globalism survey; trimmed at 4.0	numeric	0.23-4.00		

Table 2

Core Information for Each Variable in the Data File based on the YouGov Globalism Survey (yougov.rds)

Variable category	Variable name	Variable label and item wording	Variable type	Valid values	Value labels	Missing values
Descriptives	country	Participants' residency, country of data collection	character	Brazil, Canada, China, Germany, India, Indonesia, Italy, Japan, Mexico, Poland, Thailand	—	—
	n	Sample size	numeric	1,001-1,337	—	—
Climate change beliefs	ccb.main	% of people who indicated that they believe climate change is happening and mainly caused by human-activity	numeric	0-100	—	—
	ccb.part	% of people who indicated that they believe climate change is happening and partly caused by human-activity	numeric	0-100	—	—
	ccb.attr	% of people who indicated that they believe climate change is happening but not at all caused by human-activity	numeric	0-100	—	—
	ccb.trend	% of people who indicated that they believe climate change is not happening	numeric	0-100	—	—
	ccb.know	% of people who indicated that they do not know	numeric	0-100	—	—
	ccb.believer	% of ccb.main + ccb.part				

Table 3

Core Information for Each Variable in the raw Data Files (Brazil.csv, Canada.csv, China.csv, Germany.csv, India.csv, Indonesia.csv, Italy.csv, Japan.csv, Mexico.csv, Poland.csv, Thailand.csv)

Variable category	Variable name	Variable label and item wording	Variable type	Valid values	Value labels	Missing values
Collected by Qualtrics	Start date	Date and time when participant clicked on the link for the first time	character	—	—	—
	End date	Date and time when participant submitted responses	character	—	—	—
	Status	Type of response	numeric	0, 1, 2, 4, 8, 9, 12, 16, and 17	0 = IP address 1 = survey preview 2 = survey test 4 = imported 8 = spam 9 = spam preview 12 = imported spam 16 = offline 17 = offline preview	—
	Progress	Survey completion (in %)	numeric	0-100	—	—
	Duration (in seconds)	Duration of survey completion (in seconds)	numeric	≥ 0	—	—
	Finished	Survey completion (yes/no)	numeric	0, 1	0 = not finished 1 = finished	—
	RecordedDate	Date and time when Qualtrics recorded responses	character	—	—	—
	DistributionChannel	Distribution channel	character	anonymous	—	—

	UserLanguage	Language code in which participant completed survey	character	PT-BR EN FR-CA ZH-S DE HI ID IT JA ES PL TH	PT-BR = Brazilian Portuguese EN = English FR-CA = Canadian French ZH-S = Simplified Chinese DE = German HI = Hindi ID = Indonesian IT = Italian JA = Japanese ES = Spanish PL = Polish TH = Thai	—
	Q_RecaptchaScore	Google's invisible reCaptcha for bot detection	numeric	0-1	> 0.5 = participant is likely a human ≤ 0.5 = participant is likely a bot	—
Consent	Consent	Consent	numeric	1	1 = agreed	empty
Demographics I	Age (in years)	How old are you?	numeric	0, ≥ 18	0 = younger than 18	empty
	Sex	What is your sex?	numeric	0, 1	0 = female 1 = male	empty
	Citizenship	Are you a citizen of [country]?	numeric	0, 1	0 = no 1 = yes	empty
	Residency	Do you currently live in [country]?	numeric	0, 1	0 = no 1 = yes	empty
National identification	National_Identity_1	I identify as [nationality].	numeric	0-10	0 = strongly disagree 5 = neither disagree nor agree 10 = strongly agree	empty
	National_Identity_2	Being a [nationality] is an important reflection of who I am.	numeric	0-10	0 = strongly disagree 5 = neither disagree nor agree 10 = strongly agree	empty

Own and others' climate change beliefs	OwnCCBeliefs	In general, which of the following statements, if any, best describes your view?	numeric	1-5	1 = The climate is changing and human activity is mainly responsible. 2 = The climate is changing and human activity is partly responsible, together with other factors. 3 = The climate is changing but human activity is not at all responsible. 4 = The climate is not changing. 5 = I don't know.	empty
	OtherCCBeliefs_1	What percentage of [citizens], do you believe, would think the following ways about climate change? Please indicate a number from 0% (no one) to 100% (everyone) for the following statements such that they sum up to 100%. The climate is changing and human activity is mainly responsible.	numeric	0-100	—	empty
	OtherCCBeliefs_2	The climate is changing and human activity is partly responsible, together with other factors.	numeric	0-100	—	empty
	OtherCCBeliefs_3	The climate is changing but human activity is not at all responsible.	numeric	0-100	—	empty
	OtherCCBeliefs_4	The climate is not changing.	numeric	0-100	—	empty

	OtherCCBeliefs_5	Don’t know. Please indicate what percentage of [citizens] do not know what they think about climate change.	numeric	0-100	—	empty
Own and others’ biodiversity loss beliefs	OwnBLBeliefs	In general, which of the following statements, if any, best describes your view?	numeric	1-5	1 = Biodiversity is declining and human activity is mainly responsible. 2 = Biodiversity is declining and human activity is partly responsible, together with other factors. 3 = Biodiversity is declining but human activity is not at all responsible. 4 = Biodiversity is not declining. 5 = I don’t know.	empty
	OtherBLBeliefs_1	What percentage of [citizens], do you believe, would think the following ways about climate change? Please indicate a number from 0% (no one) to 100% (everyone) for the following statements such that they sum up to 100%. Biodiversity is declining and human activity is mainly responsible.	numeric	0-100	—	empty
	OtherBLBeliefs_2	Biodiversity is declining and human activity is partly responsible, together with other factors.	numeric	0-100	—	empty

	OtherBLBeliefs_3	Biodiversity is declining but human activity is not at all responsible.	numeric	0-100	—	empty
	OtherBLBeliefs_4	Biodiversity is not declining.	numeric	0-100	—	empty
	OtherBLBeliefs_5	I don't know.	numeric	0-100	—	empty
Outcomes	Discuss	How willing or unwilling are you to express your opinion on climate change among [citizens] you don't know?	numeric	1-7	1 = not at all willing 7 = very willing	empty
	Expectation_Changes_1	What percentage of [citizens], do you believe, would be willing to make the following extent of changes to how they live and work to help reduce the potential effects of climate change? Please indicate a number from 0% (no one) to 100% (everyone). No or a few changes	numeric	0-100	—	empty
	Expectation_Changes_2	Some or a lot of changes	numeric	0-100	—	empty
	Changes	How much, if anything, would you be willing to change about how you live and work to help reduce the potential effects of climate change?	numeric	1-4	1 = no changes 2 = a few changes 3 = some changes 4 = a lot of changes	empty
	Expectation_Support_1	Low or medium priority What percentage of [citizens], do you believe, think climate change should be a low/medium or high/very high priority of the government of [country]? Please indicate a number from 0% (no one) to 100% (everyone).	numeric	0-100	—	empty

	Expectation_Support_2	High or very high priority	numeric	0-100	—	empty
	Support	Do you think climate change should be a low, medium, high, or very high priority for the government of [country]?	numeric	1-4	1 = low priority 2 = medium priority 3 = high priority 4 = very high priority	empty
	Efficacy	To what extent do you think that [citizens] can jointly prevent the negative consequences of climate change?	numeric	1-7	1 = not at all 7 = very much 99 = I don't know. 999 = I don't believe in (human-caused) climate change.	empty
Comprehension	Comprehension	With this question, we would like to ensure that participants pay attention. Please select the option "Red" from the list below.	numeric	1-5	1 = Blue 2 = Red 3 = Yellow 4 = Green 5 = White	empty
Demographics II	Education	What is your highest level of completed education?	numeric	1-7	Adapted to each country 1 = lowest (no formal degree) 7 = highest (doctoral degree)	empty
	PoliticalOrientation	In politics, people sometimes talk of "left" and "right". Where would you place yourself on this scale, where 0 means the left and 10 means the right?	numeric	0-10	0 = left 10 = right	empty
	Urbanicity	Would you describe the area where you live as urban or rural?	numeric	1-3	1 = urban 2 = rural 3 = don't know/prefer not to say	empty
Metadata	Q_TotalDuration	Total duration of survey completion (in seconds)	numeric	≥ 0	—	—
	Response_Type	Response type	character	Complete, QualityFail,	Complete = complete and valid response	empty

				Screenout, Speeder	QualityFail = failed comprehension check Screenout = did not meet inclusion criteria or the respective quota was full Speeder = completed survey faster than 3 min	
	Age_Cat	Age categories (for quotas)	character	18-29 years, 30-39 years, 40-49 years, 50-59 years, 60 years or older	—	empty
	Condition / FL_54_DO	Control or intervention condition	numeric	0, 1	0 = control 1 = intervention	empty
	Comprehension_1	Whether participant passed or failed the comprehension check	character	Pass, Fail	Pass = passed check Fail = failed check	empty
Randomizer	FL11_DO	Whether participant was first presented (a) with the block on climate change vs. biodiversity loss and (b) with the block on own vs. other beliefs	character	FL_63 FL_64, FL_76, FL_88,	FL_63 = CC-BL-own-other FL_64 = BL-CC-own-other FL_76 = CC-BL-other-own FL_88 = BL-cc-other-own	empty
	randomizer_CCfirst	Whether participant was first presented with the block on climate change vs. biodiversity loss	numeric	0, 1	0 = biodiversity first 1 = climate change first	empty
	randomizer_ownfirst	Whether participant was first presented with the block on own vs. other beliefs	numeric	0, 1	0 = other beliefs first 1 = own beliefs first	empty

	OwnCCBeliefs_DO	Randomization of response options of own climate change beliefs	numeric	1|2|3|4|5 4|3|2|1|5	1|2|3|4|5 = option 1 first, option 5 last 4|3|2|1|5 = option 4 first, option 5 last	empty
	OtherCCBeliefs_DO	Randomization of response options of others' climate change beliefs	numeric	1|2|3|4|5 4|3|2|1|5	1|2|3|4|5 = option 1 first, option 5 last 4|3|2|1|5 = option 4 first, option 5 last	empty
	OwnBLBeliefs_DO	Randomization of response options of own biodiversity loss beliefs	numeric	1|2|3|4|5 4|3|2|1|5	1|2|3|4|5 = option 1 first, option 5 last 4|3|2|1|5 = option 4 first, option 5 last	empty
	OtherBLBeliefs_DO	Randomization of response options of others' biodiversity loss beliefs	numeric	1|2|3|4|5 4|3|2|1|5	1|2|3|4|5 = option 1 first, option 5 last 4|3|2|1|5 = option 4 first, option 5 last	empty
	Expectation_Changes_DO	Randomization of response options for others' changes	numeric	1|2 2|1	1|2 = option 1 first, option 2 second 2|1 = option 2 second, option 1 first	empty
	Expectation_Support_DO	Randomization of response options for others' support	numeric	1|2 2|1	1|2 = option 1 first, option 2 second 2|1 = option 2 second, option 1 first	empty

Manipulation Check

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Preregistration: Not preregistered

Project description: In this study, we tested whether participants updated their perceptions of the public consensus in response to the intervention compared to the control message, using a nationally representative sample (in terms of age and sex) from Brazil collected on *Besample*. The variables are identical to Study 1, with a few exceptions. **Table 4** lists these exceptions in the cleaned dataset *data.brazil.check.rds*.

Table 4

Core Information for the Additional Variables in the Raw and Cleaned Data File Used for the Analyses (data.brazil.check.rds)

Variable category	Variable name	Variable label and item wording	Variable type	Valid values	Value labels	Missing values
Additional variables	other.ccb.believer.pre	What percentage of [citizens], do you believe, would think the following ways about climate change? Please indicate a number from 0% (no one) to 100% (everyone). The climate is changing and human activity is mainly responsible. / The climate is changing and human activity is partly responsible, together with other factors. / The climate is changing but human activity is not at all responsible. / The climate is not changing. / Don't know. Please indicate what percentage of [citizens] do not know what they think about climate change.	numeric	0-1	—	—
	other.ccb.believer.post	What percentage of **${e://Field/Nationality_plural}**, do you believe, would think the following ways about climate change? Please indicate a number from 0% (no one) to 100% (everyone). The climate is changing and human activity is partly or mainly responsible	numeric	0-1	—	—
	att	Earlier in this study, you saw a message about a recent survey	numeric	0-100	—	Only in interven

	with information about the **percentage (%) of Brazilians** who believe that the climate is changing and human activity is partly or mainly responsible. To the best of your recollection, what was the exact number that was presented to you? Note: Please enter only the number, without the % sign.	tion; missing for control

Study 2

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Preregistration: Not preregistered

Project description: In this study, we tested the correlation between cultural tightness-looseness and pluralistic ignorance concerning willingness to contribute financially to fight climate change, using data from two cross-sectional, secondary datasets: (a) cultural tightness-looseness scores based on student and general population samples across 55 countries (*N* = 22,496; Eriksson et al., 2021) and (b) data used to calculate pluralistic ignorance scores from the same 55 countries (*N* = 60,230), selected from the Global Climate Change Survey including probability-based, nationally representative samples from 125 countries collected in the Gallup World Poll 2021/2022 (Andre et al., 2024).

Tables 5.1 and 5.2 provide core information for each variable in the data files *data_S1.csv* (i.e., data from Study 1) and *data_S2.csv* (i.e., secondary data).

Table 5.1

Core Information for Each Variable in the Cleaned Data File Used for the Analyses (data_S1.csv)

Variable category	Variable name	Variable label and item wording	Variable type	Valid values	Value labels	Missing values
Demographics	country	Name of the country	character	Brazil, Canada, China, Germany, India, Indonesia, Italy, Japan, Mexico, Poland, Thailand	—	—
Core variables	own	Actual percentage of people who believe in human-caused climate change	numeric	0-100	—	—
	others	Perceived percentage of people who believe in human-caused climate change	numeric	0-100	—	—
	ctl	Cultural tightness-looseness score	numeric	1.69-2.48	—	—

Table 5.2

Core Information for Each Variable in the Cleaned Data File Used for the Analyses (data_S2.csv)

Variable category	Variable name	Variable label and item wording	Variable type	Valid values	Value labels	Missing values
Demographics	country	Name of the country	character		—	—
Sample characteristics	n_andre	Sample size per country in Andre et al. (2024)	numeric	500-3,500	—	—
	n_eriksson	Sample size per country in Eriksson et al. (2021)	numeric	82-1,403	—	empty (*n* = 70)
Core variables	own	Actual percentage of people who would be willing to contribute financially to fight climate change	numeric	0-100	—	—
	others	Perceived percentage of people who would be willing to contribute financially to fight climate change	numeric	0-100	—	—
	ctl	Cultural tightness-looseness score	numeric	1.46-2.93	—	empty (*n* = 70)